

Leakage-Aware Multi-Task Recognition of Produce Freshness and Type for Low-Cost Mobile Deployment

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Abstract—Public fresh/rotten produce datasets are routinely reported with accuracies close to 100%. We show that, for a widely used Kaggle dataset, much of this figure is an artefact of how the data are split and how the data were collected. Its 14,682 images derive from only 1,279 source photographs taken in 35 (file-name-inferred) capture sessions. A per-file split places a sibling of 100.0% of test images in training, and an EfficientNet-B0 then scores 100.0% on freshness. Grouping copies by source photo gives 98.3%, yet 97.7% of test images still share a capture session with training. Holding out whole sessions lowers freshness accuracy to 84.9%, and expected calibration error grows from 1.4% to 12.2%. The loss is concentrated in produce types whose fresh and stale photographs came from different cameras or messaging apps: on an unseen session of stale capsicum the model is correct on only 6.9% of images, suggesting it has partly learned the acquisition source rather than freshness. On the photo-grouped split, joint training of freshness and produce type shows no difference from separate single-task models that three seeds can resolve, and MobileNetV3-Large matches EfficientNet-B0 (98.5% vs. 98.3%) at 16.7 ms CPU latency. An energy-score gate on the produce-type head rejects 97.3% of non-produce images and 86.4% of unseen produce while accepting 93.5% of in-distribution images; produce-type accuracy on an external dataset is only 62.5%. We release splits and code, and argue that shelf life and quality grade, which have no ground truth in these datasets, should not be presented as learned outputs.

Index Terms—food quality, freshness classification, multi-task learning, data leakage, out-of-distribution detection, EfficientNet, mobile deployment

I. INTRODUCTION

In informal retail settings, fruit and vegetables are usually sorted by eye, without instrumented grading. A phone photograph that flags stale produce is an attractive low-cost aid, and public image datasets of “fresh” and “rotten” or “stale” produce have made the task a popular benchmark: recent papers report accuracies close to 100% [1].

Those numbers deserve scrutiny. Several of the most-used Kaggle releases ship *pre-augmented* images: each source photograph appears many times as a rotated, flipped, translated or

noise-corrupted copy. A split drawn per *file* then places near-copies of a test image in the training set, the classic leakage failure mode described by Kaufman *et al.* [2] and surveyed across applied ML by Kapoor and Narayanan [3].

This paper studies freshness and produce-type recognition under a protocol that removes this leakage, and reports what remains. Our contributions are:

- 1) A **leakage audit** of the Kaggle “Fresh and Stale Images of Fruits and Vegetables” dataset [4]: its 14,682 images collapse into 1,279 near-duplicate clusters (1,622 by filename alone), and a per-file split puts a sibling of 100.0% of test images into training. A second, held-out-session analysis shows that the remaining in-domain accuracy partly rests on capture-session cues. We release both splits.
- 2) A controlled comparison of a **multi-task** network (freshness + produce type) against single-task networks with identical backbones, and of EfficientNet-B0 [5] against MobileNetV3-Large [6], over three seeds, with a group-bootstrap confidence interval for a single run.
- 3) An evaluation of **post-hoc out-of-distribution (OOD) gates** for rejecting non-produce and unseen-produce photos, against an external dataset and CIFAR-10.
- 4) An honest deployment design: attributes that have no ground truth in any available dataset (quality grade, remaining shelf life) are *not* learned; they are reported as explicitly labelled heuristics.

II. RELATED WORK

Freshness classification. CNN and transformer classifiers trained on public fresh/rotten datasets report accuracies in the high 90s. Transfer-learned AlexNet reaches 99.8% on a public fresh/rotten dataset [7], and a multi-task ResNet ensemble predicts freshness and fruit type jointly (98.5% and 97.4%) [8]. Closest to our setting, Multi-Task NoisyViT [1] reports 99.85% on the same Fresh/Stale dataset we use, with random 60/20/20

splits; the only deduplication it describes is MD5 hashing, applied when building its merged Freshness44 set. In [7] the data are rotation-augmented *before* a random 80/10/10 split. None of these works groups augmented or near-duplicate copies of a source photo before splitting, and we found no freshness study that measures the resulting leakage.

Shelf-life estimation. Predicting remaining shelf life requires measured end points. Datasets that follow produce over time until spoilage exist [9], and the recent CNN+DeiT study of [10] forecasts spoilage with a symmetric mean absolute percentage error above 40%. The dataset we use has no such measurements, so we treat shelf life as out of scope for learning.

Multi-task learning. Sharing a representation across related tasks can act as an inductive bias [11], [12]; whether it helps depends on task relatedness, which we test directly.

OOD detection. Maximum softmax probability (MSP) [13] and the energy score [14] are standard post-hoc detectors that need no retraining, which suits on-device use.

Leakage. Kaufman *et al.* [2] formalise leakage; Kapoor and Narayanan [3] document its effect on published results across fields, including duplicated samples across splits.

III. DATA AND LEAKAGE AUDIT

A. Datasets

Main set. The Kaggle “Fresh and Stale Images of Fruits and Vegetables” dataset [4] contains 14,682 images of six produce types (apple, banana, bitter melon, capsicum, orange, tomato), each labelled fresh or stale by folder. Both labels are used as ground truth.

External set. The Kaggle “Fruit and Vegetable Image Recognition” dataset [15] (36 classes, no freshness labels) is used only for evaluation: its apple, banana, orange, tomato, capsicum and bell-pepper images (590) form a cross-dataset produce-type test, and its 3,206 images of other classes form a near-OOD set. We drop 29 external images that near-duplicate a main-set image.

Far-OOD set. 2,000 randomly drawn CIFAR-10 test images [16], upsampled to the model input size.

B. Leakage and the grouped split

File names in the main set encode the augmentation applied to a source photo (prefixes such as `rotated_by_15_`, `saltandpepper_`, `translation_`, `vertical_flip_` and `Copy of`, and suffixes such as `.jpg_0_4456.jpg`). We normalise each name back to its source and join files with the same source, within a produce type, into one group (1,622 groups). We then compute a 64-bit difference hash of every image and merge groups whose images lie within a Hamming distance of 2 bits, again only within a produce type (low-texture images of different produce otherwise collide). This yields 1,279 clusters.

Because merging is transitive, a single chance match can join distinct photographs: 18 clusters (3,883 images) contain both fresh and stale images, and the largest has 933 images. Over-merging cannot leak test images into training; it only

TABLE I
GROUPED SPLIT: IMAGES PER PRODUCE TYPE, FRESHNESS AND SPLIT.

Type	Train		Val		Test	
	Fresh	Stale	Fresh	Stale	Fresh	Stale
Apple	1,165	1,741	300	307	228	294
Banana	1,219	1,036	158	223	204	208
Bitter melon	246	294	41	0	40	63
Capsicum	836	860	31	41	123	0
Orange	1,033	1,144	228	192	205	259
Tomato	941	736	40	82	0	164

TABLE II
ACCURACY (%) UNDER A PER-FILE SPLIT VERSUS THE GROUPED SPLIT. Δ = INFLATION CAUSED BY LEAKAGE.

Model	Freshness			Produce type		
	Per-file	Grouped	Δ	Per-file	Grouped	Δ
HSV hist.+LR	94.0	86.3	+7.7	96.6	80.0	+16.6
B0 multi-task	100.0	98.3	+1.7	100.0	99.8	+0.2

makes the split coarser. We therefore call the units “clusters” rather than photographs.

The grouped split assigns whole clusters to train, validation and test with stratified group k -fold sampling. Because some photo sessions contain hundreds of copies, we select, among 50 candidate seeds, the split whose per-stratum validation and test shares are closest to 15% (seed 49); the result has 11,251/1,643/1,788 images and 200 independent test groups (Table I). For comparison we also draw a per-file split with the same proportions. Under it, 100.0% of test images have a group sibling in training.

IV. METHOD

A. Model

A timm [17] backbone pretrained on ImageNet [18] (EfficientNet-B0 or MobileNetV3-Large) produces a pooled feature vector. Each task has its own head (dropout 0.3, linear 256, ReLU, dropout 0.2, linear to the task’s classes): freshness (2 classes) and produce type (6 classes). The multi-task loss is the mean of the two cross-entropy terms; single-task baselines use the same backbone and head with one task.

B. Heuristic attributes

No available dataset contains graded quality or measured shelf life, so neither is learned. For the user interface only, the service reports a grade from the predicted $p = P(\text{fresh})$ (A if $p \geq 0.85$, B if $p \geq 0.5$, else C) and an indicative shelf life $d_t p$, where d_t is a room-temperature reference for produce type t . Both fields are flagged as heuristic in the API and the app, and we make no accuracy claim about them.

C. OOD gate

We compare four post-hoc scores: negative entropy of the freshness head (the gate used by the previous version of the system), MSP of the freshness head, MSP of the produce-type head [13], and the energy score (negative free energy at

TABLE III

FRESHNESS ACCURACY (%; POOLED OVER THREE SEEDS) PER PRODUCE/FRESHNESS STRATUM: PHOTO-GROUPED TEST (SESSIONS SEEN IN TRAINING) VERSUS HELD-OUT-SESSION TEST (AN UNSEEN CAPTURE SESSION). n : HELD-OUT TEST IMAGES; $-$: STRATUM ABSENT FROM THE PHOTO-GROUPED TEST.

Stratum	Held-out session	n	Grouped	Held-out
Apple (stale)	Screenshots 2018-06-08	975	98.9	97.4
Bitter gourd (fresh)	Bitter-gourd 1 series	41	100.0	99.2
Bitter gourd (stale)	IMG 2020-08-25	84	100.0	100.0
Capsicum (fresh)	Day series	124	100.0	26.6
Capsicum (stale)	IMG 2020-09-01	164	—	6.9
Orange (fresh)	Screenshots 2018-06-12	482	94.6	99.9
Tomato (fresh)	IMG 2020-07-25	122	—	55.2
Tomato (stale)	IMG 2020-07-29	163	100.0	99.0

temperature $1) \log \sum_k e^{z_k}$ of the produce-type logits [14]. The deployed gate uses the energy score, with the threshold set to accept 95% of validation images; the threshold is stored with the checkpoint.

D. Training and evaluation protocol

Images are resized to 224×224 and normalised with ImageNet statistics. Training augmentation: random resized crop (scale 0.8–1), horizontal and vertical flips, 90° rotations, colour jitter, Gaussian noise and coarse dropout. We train with AdamW [19] (learning rate 3×10^{-4} , weight decay 10^{-4} , batch 64), one epoch at 10% of the base learning rate followed by cosine decay, mixed precision, gradient clipping at 1.0, up to 10 epochs with early stopping on validation loss (patience 3); runs stopped after 5–10 epochs. Each configuration is trained with seeds 0, 1 and 2 and we report mean $\pm$ standard deviation; calibration is measured with the expected calibration error (ECE, 15 bins) [20]. For a single run we also report a 95% confidence interval from 1,000 bootstrap resamples of *test groups* [21], since images within a group are not independent. As a floor we train a logistic-regression classifier on $8 \times 8 \times 8$ HSV colour histograms.

Hardware: one NVIDIA RTX 4060 Laptop GPU (8 GB) for training; latency is measured on the Intel Core i7-14700HX CPU with 4 threads, batch size 1, as the median of 200 runs interleaved across all models. Software: Python 3.13, PyTorch 2.12, timm 1.0.24, PyTorch Lightning 2.6.1, Albumentations 2.0.8. Code, split files and per-run logs are available at https://github.com/mathuryashash/FreshTrack_AI.

V. RESULTS

A. Leakage from augmented copies

Table II and Fig. 1 compare the per-file and grouped protocols. Under the per-file split every EfficientNet-B0 seed classifies every test image correctly (100.0% freshness, 100.0% type). The grouped split lowers this to 98.3% and 99.8%. The effect is larger for the colour-histogram baseline, whose freshness accuracy drops from 94.0% to 86.3% and type accuracy from 96.6% to 80.0%: the per-file split lets even a model with no spatial features look almost as good as a deep

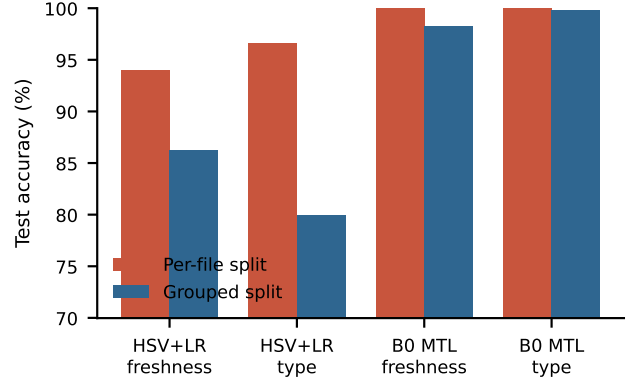


Fig. 1. Test accuracy under the per-file and grouped splits.

network, which hides how much of the task the CNN actually solves.

B. Session-level dependence

Grouping by source photo does not make test images independent of training. File names indicate only 35 capture sessions (a capture date or camera series within a class; we infer them from file-name patterns), and 97.7% of grouped-split test images come from a session that also contributes training images; 4 of the 12 produce/freshness strata consist of a single session. For the 8 strata with more than one session we built a held-out-session split (10,310/2,050/2,155 images) in which each such stratum contributes one entire, unseen session to the test set. We drop the 167 training images that are augmented copies (same file-name source) or near-duplicates (dHash within 2 bits) of a test image; no remaining training image is within 2 bits of a test image. Clusters from the grouped split may still span train and test here, because chained merging links different photographs of the same session, which is exactly the dependence this protocol probes.

On this split freshness accuracy is 84.9% (vs. 98.3% on the grouped split), while produce-type accuracy stays at 98.5%, and the freshness ECE rises from 1.4% to 12.2%: the model becomes confidently wrong. Table III breaks this down by stratum. On the 6 strata that also occur in the grouped test, accuracy falls only from 98.9% to 87.0%; apple, bitter gourd and orange sessions transfer almost perfectly. The failures are capsicum (26.6% on an unseen fresh session, 6.9% on an unseen stale session) and fresh tomato (55.2%). In both cases the held-out session’s capture source appears in training only with the opposite label: every stale capsicum left in training is a WhatsApp-compressed photograph, while every camera (IMG-series) capsicum in training is fresh, so camera photographs of stale capsicums are called fresh; likewise, all IMG-series tomatoes in training are stale, and the held-out IMG-series fresh tomatoes are often called stale. This pattern is consistent with the network having partly learned the acquisition source, a shortcut that per-file and per-photo

TABLE IV

TEST RESULTS ON THE PHOTO-GROUPED SPLIT (CAPTURE SESSIONS ARE STILL SHARED WITH TRAINING), MEAN $\pm$ STD OVER THREE SEEDS (%). CPU LATENCY: BATCH 1, 4 THREADS, MEDIAN OF 200 INTERLEAVED RUNS.

Model	Fresh acc.	Fresh macro-F1	Type acc.	Type macro-F1	Params (M)	CPU (ms)
EfficientNet-B0, multi-task	98.3 $\pm$ 0.1	98.3 $\pm$ 0.1	99.8 $\pm$ 0.2	99.8 $\pm$ 0.2	4.67	22.3
EfficientNet-B0, freshness only	97.6 $\pm$ 0.4	97.6 $\pm$ 0.4	–	–	4.34	24.6
EfficientNet-B0, type only	–	–	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	4.34	22.1
MobileNetV3-L, multi-task	98.5 $\pm$ 0.4	98.5 $\pm$ 0.4	99.7 $\pm$ 0.4	99.8 $\pm$ 0.2	4.86	16.7
HSV histogram + LR	86.3	86.2	80.0	76.0	–	–

TABLE V

OUT-OF-DISTRIBUTION DETECTION (% , MEAN $\pm$ STD OVER THREE SEEDS). NEAR-OOD: UNSEEN PRODUCE CLASSES FROM AN EXTERNAL DATASET; FAR-OOD: CIFAR-10 TEST IMAGES. HIGHER AUROC AND LOWER FPR@95%TPR ARE BETTER.

Score	Near-OOD (3,206 images)		Far-OOD (2,000 images)	
	AUROC	FPR@95	AUROC	FPR@95
<i>EfficientNet-B0, multi-task</i>				
Neg. entropy, freshness (v1 gate)	91.0 $\pm$ 2.5	37.7 $\pm$ 8.7	95.8 $\pm$ 0.8	25.8 $\pm$ 7.8
MSP, freshness	91.1 $\pm$ 2.5	37.7 $\pm$ 8.7	95.8 $\pm$ 0.8	25.8 $\pm$ 7.8
MSP, produce type	97.8 $\pm$ 0.4	8.8 $\pm$ 1.8	98.9 $\pm$ 0.4	3.7 $\pm$ 2.6
Energy, produce type	98.2 $\pm$ 0.4	7.7 $\pm$ 2.0	99.4 $\pm$ 0.2	2.7 $\pm$ 1.5
<i>MobileNetV3-L, multi-task</i>				
Neg. entropy, freshness (v1 gate)	91.3 $\pm$ 1.6	42.9 $\pm$ 4.6	95.1 $\pm$ 1.5	32.3 $\pm$ 11.8
MSP, freshness	91.3 $\pm$ 1.6	42.9 $\pm$ 4.6	95.1 $\pm$ 1.5	32.3 $\pm$ 11.8
MSP, produce type	96.0 $\pm$ 1.2	19.1 $\pm$ 7.2	98.3 $\pm$ 0.6	6.8 $\pm$ 4.4
Energy, produce type	96.6 $\pm$ 1.6	15.4 $\pm$ 8.4	99.3 $\pm$ 0.5	3.6 $\pm$ 3.1

splits cannot reveal because 97.7% of their test images share a capture session with training images.

C. Multi-task versus single-task learning and backbones

Table IV reports the grouped-split results. The multi-task EfficientNet-B0 reaches 98.3% freshness and 99.8% type accuracy, against 97.6% and 100.0% for the single-task networks. The differences are below one percentage point and point in opposite directions (freshness favours the multi-task model, type the single-task one); three seeds cannot resolve them, so we find no evidence that joint training helps or hurts, while one shared network halves inference cost. MobileNetV3-Large matches EfficientNet-B0 (98.5% and 99.7%) with 4.86M parameters and 16.7ms CPU latency against 22.3ms, and is the model we deploy. For a single seed-0 run the 95% group-bootstrap interval of freshness accuracy is [96.4, 99.6]%, reflecting that the 200 test groups, not the 1,788 images, are the independent units.

D. Cross-dataset produce type

On 590 external images of five of the six produce types (bell pepper is mapped to capsicum; there is no bitter gourd), type accuracy is only 62.5% (MobileNetV3) and 63.3% (EfficientNet-B0), compared with about 99% in-domain. The external set has no freshness labels; MobileNetV3 calls 83.0% of these retail-style photos fresh and EfficientNet-B0 72.9%, which is plausible but not an accuracy.

E. Out-of-distribution gate

Table V compares post-hoc scores. Scores computed on the six-way produce-type head separate in- from out-of-distribution images much better than scores on the binary

freshness head (near-OOD AUROC 96.6% for energy on the type head vs. 91.3% for MSP on the freshness head). For a two-class head, entropy is a monotone function of MSP, so entropy ranks inputs exactly like MSP. The previous version of the system rejected an input when freshness entropy exceeded 1.5 bits or its maximum probability fell below 0.3; for a two-class head neither condition can occur (entropy is at most 1 bit and the maximum probability at least 0.5), so that gate never fired. With the threshold set at 95% validation TPR, the MobileNetV3 energy gate, averaged over three seeds, accepts 93.5% of in-distribution test images and rejects 97.3% of CIFAR-10 images but only 86.4% of unseen produce (77.1–91.4% across seeds; the deployed seed: 94.5%, 94.2% and 77.1%). A photo of an unsupported vegetable is the hard case.

VI. DEPLOYMENT

The model is served by a FastAPI service that validates uploads (content type, size, file signature, image decoding), applies EXIF orientation, runs the model, applies the OOD gate, and logs predictions and optional user corrections to SQLite for later relabelling. A Flutter client displays the freshness and produce-type predictions, labels the grade and shelf-life fields as estimates, and avoids food-safety wording (“looks fresh” rather than “safe to eat”). The checkpoint travels with a metadata file holding the label order, preprocessing constants and OOD threshold, so a client cannot silently use different preprocessing. We deploy the MobileNetV3-Large multi-task seed with the lowest validation loss. With 4.86M parameters and 16.7ms CPU latency, on-device inference is feasible; exporting it to ONNX Runtime for offline use on phones is future work.

VII. LIMITATIONS

Capture sessions are inferred from file-name patterns, and hash-based clustering over-merges some distinct photographs, so both units are approximations. The held-out-session split covers only the 8 strata that have more than one capture session; for the others no session-independent estimate exists, so grouped-split numbers remain optimistic upper bounds. Where fresh and stale images of a produce type were captured with different devices or apps, freshness is confounded with acquisition source; we can expose this confound but not remove it without new data. Freshness labels are the dataset authors' binary folder labels, without inter-rater agreement or intermediate ripeness stages. Most images show single items on plain backgrounds, and the low cross-dataset accuracy suggests performance on cluttered market photos will be lower still; we did not collect field images. CIFAR-10 images are low-resolution and upsampled, which makes them easy far-OD examples. Latency was measured on a laptop CPU, not a phone. Finally, the quality grade and shelf-life values shown to users are heuristics without any validation.

VIII. CONCLUSION

Near-perfect accuracy on public fresh/stale produce benchmarks largely reflects the evaluation protocol. Removing augmented copies changes little for a CNN but a lot for a weak baseline, and holding out capture sessions reveals a drop in freshness accuracy, consistent with an acquisition-source shortcut, that per-image or per-photo splits cannot show. On the photo-grouped split, one multi-task MobileNetV3-Large recognises produce type and freshness as well as separate EfficientNet-B0 models at lower cost, and an energy gate on its produce-type head rejects most non-produce inputs. Progress on this problem now depends more on data than on architectures: multi-session, multi-location captures with time-to-spoilage measurements are needed before shelf-life prediction from RGB images can be evaluated at all.

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